

User Manual

MidiShip-M — Motorized MIDI Controller with 4 Faders

1. Introduction

MIDI Ship is a professional MIDI controller featuring four motorized faders, touch sensors, an OLED display, an encoder with a push button, and built-in Wi-Fi. The device supports three main operating modes:

- **CC Normal** — MIDI Control Change (CC) messages (7-bit or 14-bit) with up to 10 user-definable presets.
- **CC Edit** — editing CC number assignments for faders and the encoder.
- **HUI** — HUI protocol emulation for controlling DAWs that support this protocol (e.g., Pro Tools, Logic Pro, Cubase, and others).

Additionally, a **Service Mode** is available for motor parameter adjustment, preset reset, Wi-Fi management, and other system functions.

The controller connects via USB (class-compliant MIDI device) and can also work over Wi-Fi with a built-in web interface for remote configuration.

2. Package Contents & Appearance

- 4 motorized faders (100 mm travel) with touch sensors.
 - Encoder with push button (press, rotate, hold).
 - OLED display 128×64.
 - Two USB-C ports:
 - One for logic power and MIDI connection to the computer.
 - The second for motor power (recommended power supply: 5 V, ≥ 2 A).
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3. Connection & Setup

USB Connection

Connect the first USB-C port (main) to your computer or USB hub — this provides logic power and MIDI communication. The controller will be automatically recognized as a MIDI device (no drivers required). In your DAW, select the "MidiShip-M" input/output port.

For motor operation, additionally connect the second USB-C port to a power supply providing 5 V and at least 2 A (recommended). This ensures stable motor performance without power drops.

Power Supply

- Logic and display are powered from the first port (500 mA is sufficient).
- Motors require the second port (≥ 2 A).

Power-On

After applying power to the first port, the screen will show the "MIDI Ship" splash screen, then the controller will enter the last saved mode (CC or HUI).

4. Main Controls

Faders

- Moving a fader sends MIDI messages (CC or HUI volume).
- When a fader is touched (touch sensor), "catch" mode is activated — the motor disengages, and the controller starts transmitting data from that fader.
- After releasing the fader, the motor returns it to the target position (if it has changed externally).

Encoder

- Rotation — changes values (in CC mode) or navigates menus/banks (in HUI mode).
- Short press (1 click) — confirm, switch modes, or perform actions.
- Hold — enter edit mode (CC Edit) or reset parameters.
- Multiple clicks (2, 3, 4, 5, 10) — perform special commands (see below).

OLED Display

- Displays the current preset, fader values (7-bit), CC numbers, HUI channel states (mute/solo), transport, banks, and more.
 - After 10 minutes of inactivity, the display automatically turns off to save power; it wakes up when a fader is touched or the encoder is rotated.
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5. Operating Modes

5.1. CC Normal Mode

- **Purpose:** Control MIDI Continuous Controller (CC) messages.

- **Display:** Shows the preset number (**Preset: X**), fader values (0–127), and their assigned CC numbers.
- **Faders:** Moving a fader sends CC commands with the number specified in the preset.
- **Encoder:**
 - Rotation — changes the CC value and sends MIDI (CC number is set in the current preset for the encoder).
 - 1 click or clockwise rotation with the button held — switch to the next preset.
 - 4 clicks or counter-clockwise rotation with the button held — switch to the previous preset.
 - Hold — enter CC Edit mode.
- **If Send All CC is enabled:** A single click does **not** switch presets but sends all current fader and encoder values to the DAW using their assigned CC numbers.
- **Quick clicks:**
 - 3 clicks — switch to HUI mode.
 - 5 clicks — toggle extended MIDI channel display (**Ch: X**) next to the preset number.
 - 10 clicks — enter Service Mode.

5.2. CC Edit Mode

- Allows reassigning CC numbers for faders and the encoder.
- **Entry:** Hold the encoder in CC Normal mode.
- **Encoder:**
 - Rotation — changes the CC number assigned to the encoder.
 - 1 click or clockwise rotation with the button held — switch to the next preset.
 - 4 clicks or counter-clockwise rotation with the button held — switch to the previous preset.
 - 2 clicks — return to CC Normal without saving changes.
 - 3 clicks — select the next MIDI channel for this preset (extended mode only).
 - Hold — saves presets and returns to CC Normal (the controller will remain in CC Normal after power-on).

5.3. HUI Mode

- **Purpose:** DAW control via the HUI protocol.
- **Display:** Shows "HUI Mode", fader bank (1–8), channels (1–4, 5–8). Displays volume values (in HUI format: signed numbers), mute/solo states, fader touch, and transport controls.
- **Faders:**
 - Touch sends a channel capture message.
 - Movement sends volume value (14-bit) for the current channel.
 - After release, the fader follows the position from the DAW.
- **Encoder:**
 - Left/right rotation with button held — switch fader banks (halves).
 - 1 click — mute selected channel.
 - 2 clicks — solo selected channel.

- 3 clicks — switch to CC Normal mode.
- 4 clicks — toggle Flip mode (Volume / Send). In Flip mode, the encoder switches the Send number (a–e).
- Hold — saves HUI mode (the controller will remain in HUI after power-on).
- **Transport:** If Flip is inactive, encoder rotation controls Play/Stop (right/left).

5.4. Service Mode

Entry:

- Method 1: Press and hold the encoder button during power-on until the splash screen appears, then release.
- Method 2: In any mode (except CC Edit), perform 10 clicks of the encoder.

Menu items (navigation — rotate encoder, select — click):

- **Info** — firmware version, board revision, IP address (if Wi-Fi is connected).
- **Reset cur** — reset the current preset to factory defaults.
- **Reset all** — reset all presets to factory defaults.
- **Reset Wi-Fi** — reset Wi-Fi settings and restart (the controller will enter access point mode).
- **Vegas** — motor demonstration. Rotation — change modes, click — exit.
- **Motor's settings** — motor parameter adjustment (see Section 6).
- **HUI Ping** — enable/disable HUI ping packets (required for some DAWs).
- **14-bit CC** — switch CC format: 7-bit or 14-bit (for faders in CC Normal).
- **Send All CC** — enable/disable sending all fader and encoder values of the current preset with a single encoder click in CC Normal mode.
- **Motors** — enable/disable fader motors.
- **Identity Reply** — allow response to SysEx Identity Request. Some DAWs (e.g., Cubase) may request device identification.
- **Exit** — exit to the last active mode.

6. Motor Settings

The Service Menu contains a submenu for fine-tuning motor behavior. The following parameters are available:

Parameter	Description	Range
Speed min	Minimum speed at which the motor starts moving (higher = sharper start).	100–250
Speed max	Maximum motor speed.	100–255

Dist max	Distance (in ADC units) at which the motor reaches maximum speed.	100–1000
Dist Stop	Distance at which the motor stops (considers the target reached). Lower value = more accurate stop, but may cause oscillations.	1–2000
Dead Zone	ADC noise threshold in the bottom position (values below the threshold are forced to 0).	10–100 0

Parameter adjustment is done by rotating the encoder; selecting an item is done by clicking. Holding the encoder for a long time resets all parameters to factory defaults.

7. Wi-Fi & Web Interface

7.1. Initial Wi-Fi Connection

On first power-on (or after a Wi-Fi reset), the controller creates its own access point named **MidiShip-AP** (no password).

1. Connect to this network from your computer or smartphone.
2. In your browser, go to **192.168.4.1**.
3. The "MidiShip Wi-Fi Configuration" page will open. Available Wi-Fi networks will be scanned.
4. Select your home network from the list or enter the SSID manually.
5. Enter the password and click **Connect**.
6. The controller will connect to your network, save the settings, and restart.

7.2. Accessing the Web Interface After Connection

After restart, the controller will receive an IP address on your local network. You can find the IP address:

- In the **Info** section of the Service Menu.
- In your router's settings (list of connected devices).

Enter this IP address in your browser — the main control page will open.

7.3. Web Interface Features

The main page contains:

- **Preset Table** (10 presets) — edit CC numbers for each fader, the encoder, and the MIDI channel for each preset.
- **Motor Settings** — same parameters as in the Service Menu.
- **Options:**

- **HUI Ping** — enable/disable HUI ping packets.
- **14-bit CC** — switch CC format (7 or 14 bits).
- **Send All CC** — enable sending all fader and encoder values on click.
- **Motors On** — global motor enable/disable.
- **Identity Reply** — allow response to Identity Request.
- **Buttons:**
 - **Save All Settings** — save all changes to non-volatile memory.
 - **Firmware Update** — go to the firmware upload page.
 - **Export Settings** — export all settings as a JSON file.
 - **Import Settings** — import settings from a previously exported file (the controller will restart after import).
 - **Wi-Fi Reset** — reset Wi-Fi settings and restart (you will need to reconfigure the connection).
 - **Factory Reset** — full reset of all settings to factory defaults (including presets, motors, flags).

7.4. Firmware Update

1. Open the web interface and click **Firmware Update**.
2. On the page that appears, select the firmware file (**.bin** extension).
3. The update process will start automatically.
4. Wait for the progress bar to complete. The controller will reboot with the new firmware.



Important:

Ensure a stable connection to your access point.

Do not power off or disconnect USB during the update!

8. Special Features

8.1. Extended Mode

5 clicks of the encoder in CC mode toggle the display of the MIDI channel number next to the preset number. The channel can be changed in CC Edit or via the web interface.

8.2. HUI Emulation

- Fader banks (1–8) are supported, divided into two halves (1–4 and 5–8). Bank switching is done by rotating the encoder left/right while holding the button in normal mode (without Flip).
- **Flip** (Send mode): 4 clicks of the encoder. In this mode, the faders control Send levels (a–e), and the encoder switches the Send number.

8.3. Touch Sensors

Each fader is equipped with a capacitive touch sensor. When touched:

- In HUI: A channel capture message is sent, the motor disengages, and data is transmitted.

8.4. Send All CC

In CC Normal mode, the **Send All CC** function allows you to send the current values of all faders and the encoder of the current preset to the DAW with a single encoder click. This is useful for quickly synchronizing the controller state with the software — for example, after opening a new project, changing a preset, or when desynchronization occurs.

How it works:

- When Send All CC is enabled (in the Service Menu or web interface), a short press (1 click) of the encoder in CC Normal mode stops switching presets and instead sends all current fader and encoder values.
- Each message contains the CC number assigned to that element in the current preset and its current value (0–127).
- Sending happens sequentially (fader 1, 2, 3, 4, and the encoder), but fast enough for the DAW to perceive it as an instant "snapshot" of the controller's state.

When to use:

- At session start, to align virtual parameters in the DAW with the actual positions of the faders.
- After loading a new preset, to instantly update all assignments in the DAW.
- If synchronization is lost for any reason (e.g., after USB reconnection), you can press Send All and restore control without manually moving each fader.

Important note:

In some DAWs (e.g., in "pickup" or "takeover" modes), parameters do not change until the physical controller "catches up" to the current value in the software. In such cases, pressing Send All may not have the expected effect, as the DAW expects physical movement of the element. Make sure that your DAW is set to a mode where incoming MIDI messages are accepted immediately (e.g., "absolute" or "direct").

Managing the option:

Send All CC can be enabled/disabled via the Service Menu (item "Send All CC") or via the web interface (checkbox "Send All CC" in the Options section). The state is stored in the controller's memory.

8.5. Motor Power from the Second USB Port

For motor operation, 5–9 V is sufficient. If your USB port or hub provides ≥ 1.5 A, an external adapter is not required. Note: if you supply more than 5 V, you will need to readjust the minimum and maximum speed (Speed min/max) in the motor settings menu.

8.6. Global Motor Control

Motors On/Off — allows you to disable all motors. When disabled, the motors stop and will not start again until reactivated. This function is available both in the Service Menu and via the web interface.

8.7. Identity Reply

Identity Reply — controls the response to the standard MIDI identification request (SysEx Identity Request). Some DAWs use this request for automatic controller recognition. By default, this function is disabled. In rare cases, enabling it may help speed up DAW startup. For example, if you are using Cubase on Windows 11 and experience freezes during MIDI initialization, it is recommended to ensure that Identity Reply is enabled (ON), as Cubase expects a response from the device.

9. Saving Settings

All changes (presets, motor settings, MIDI channels, flags) are automatically saved to non-volatile memory (Preferences) when exiting edit mode or when saving via the web interface. The current preset and mode are remembered and restored on power-on.

10. Troubleshooting

Problem	Likely Cause	Solution
Controller not recognized as a MIDI device	Driver not installed (rare), USB cable	Try a different USB cable, try another port. Windows may require a driver (not included).
Faders do not move or move jerkily	Insufficient motor power, USB cable	Ensure the second USB-C port is connected to a power supply with ≥ 2 A. Try a USB-C to USB-A cable or adapter.
Display turns off	Power-saving feature activated	Touch any fader or rotate the encoder — the display will wake up.
Cannot connect to Wi-Fi	Incorrect password, or 5 GHz network (controller only supports 2.4 GHz)	Check the password, try a 2.4 GHz network.
Flickering CC or HUI values on the screen	Dead Zone value too low	Increase Dead Zone in motor settings.

Motors "jitter" at the zero position	Distance Stop value too small	Increase Distance Stop in motor settings.
Motors do not reach the target, do not respond to smooth changes, overshoot, etc.	Motor power supply voltage	Change the motor power supply. Experiment with motor settings.

11. Technical Specifications (Brief)

- **Faders:** 4 (100 mm, motorized, with touch)
 - **Encoder:** with push button (24 pulses)
 - **Display:** OLED 128×64
 - **MIDI:** USB (MIDI class)
 - **Protocols:** CC (7/14-bit), HUI
 - **Wi-Fi:** 802.11 b/g/n (2.4 GHz), access point or client
 - **Power:** 5 V USB (two ports: logic — up to 500 mA, motors — ≥ 2 A)
 - **Dimensions:** 180 × 85 × 40 mm
 - **Weight:** approx. 600 g
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12. Contacts & Support

- **Website:** www.midiship.com
- **Firmware:** available on GitHub <https://github.com/MidiShip/MidiShip-M>
- **Support:** contact@midiship.com

Thank you for choosing MIDI Ship! Good luck and creative success!